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Probability of Transition to Psychosis in Individuals at Clinical High Risk
An Updated Meta-analysis

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IMPORTANCE Estimating the current likelihood of transitioning from a clinical high risk for psychosis (CHR-P) to psychosis holds paramount importance for preventive care and applied research.

OBJECTIVE To quantitatively examine the consistency and magnitude of transition risk to psychosis in individuals at CHR-P.

DATA SOURCES PubMed and Web of Science databases until November 1, 2020. Manual search of references from previous articles.

STUDY SELECTION Longitudinal studies reporting transition risks in individuals at CHR-P.

DATA EXTRACTION AND SYNTHESIS Meta-analysis compliant with Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) and Meta-analysis of Observational Studies in Epidemiology (MOOSE) reporting guidelines; independent data extraction, manually and through digitalization of Kaplan-Meier curves.

MAIN OUTCOME AND MEASURES Primary effect size was cumulative risk of transition to psychosis at 0.5, 1, 1.5, 2, 2.5, 3, 4, and more than 4 years' follow-up, estimated using the numbers of individuals at CHR-P transitioning to psychosis at each time point. These analyses were complemented by meta-analytical Kaplan-Meier curves and speed of transition to psychosis (hazard rate). Random-effects meta-analysis, between-study heterogeneity analysis, study quality assessment, and meta-regressions were conducted.

RESULTS A total of 130 studies and 9222 individuals at CHR-P were included. The mean (SD) age was 20.3 (4.4) years, and 5100 individuals (55.3%) were male. The cumulative transition risk was 0.09 (95% CI, 0.07-0.10; k = 37; n = 6485) at 0.5 years, 0.15 (95% CI, 0.13-0.16; k = 53; n = 7907) at 1 year, 0.20 (95% CI, 0.17-0.22; k = 30; n = 5488) at 1.5 years, 0.19 (95% CI, 0.17-0.22; k = 44; n = 7351) at 2 years, 0.25 (95% CI, 0.21-0.29; k = 19; n = 3114) at 2.5 years, 0.25 (95% CI, 0.22-0.29; k = 29; n = 4029) at 3 years, 0.27 (95% CI, 0.23-0.30; k = 16; n = 2926) at 4 years, and 0.28 (95% CI, 0.20-0.37; k = 14; n = 2301) at more than 4 years. The cumulative Kaplan-Meier transition risk was 0.08 (95% CI, 0.08-0.09; n = 4860) at 0.5 years, 0.14 (95% CI, 0.13-0.15; n = 3408) at 1 year, 0.17 (95% CI, 0.16-0.19; n = 2892) at 1.5 years, 0.20 (95% CI, 0.19-0.21; n = 2357) at 2 years, 0.25 (95% CI, 0.23-0.26; n = 1444) at 2.5 years, 0.27 (95% CI, 0.25-0.28; n = 1029) at 3 years, 0.28 (95% CI, 0.26-0.29; n = 808) at 3.5 years, 0.29 (95% CI, 0.27-0.30; n = 737) at 4 years, and 0.35 (95% CI, 0.32-0.38; n = 114) at 10 years. The hazard rate only plateaued at 4 years' follow-up. Meta-regressions showed that a lower proportion of female individuals (β = −0.02; 95% CI, −0.04 to −0.01) and a higher proportion of brief limited intermittent psychotic symptoms (β = 0.02; 95% CI, 0.01-0.03) were associated with an increase in transition risk. Heterogeneity across the studies was high (I^2 range, 77.91% to 95.73%).

CONCLUSIONS AND RELEVANCE In this meta-analysis, 25% of individuals at CHR-P developed psychosis within 3 years. Transition risk continued increasing in the long term. Extended clinical monitoring and preventive care may be beneficial in this patient population.

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Supplemental content

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Indicated prevention in individuals at clinical high risk of psychosis (CHR-P)[1] encompasses detection, prognosis, and intervention, which have been recently appraised by an umbrella review.[2] The recruitment of young individuals at CHR-P typically leads to them accumulating risk factors for psychosis[3-5] that are associated with functional impairments[6] and attenuated psychotic symptoms.[7] These individuals seek intervention[8,9] at specialized mental health clinics,[10] where psychometric instruments are used to formulate a group-level prognosis (ie, at risk or CHR-P vs not at risk or not CHR-P; area under the receiver operating characteristic curve, 0.9 at 38 months[11]). Transition to psychosis is the primary outcome in this field, and it is associated with several clinically meaningful real-world outcomes.[12] Therefore, prognostication[13] of transition to psychosis in individuals at CHR-P holds paramount importance for clinical care.

Risk of transition to psychosis in individuals at CHR-P as estimated by our earlier meta-analysis (including studies until January 2011) was 18% at 6 months, 22% at 1 year, 29% at 2 years, and 36% at 3 years.[14] After a decade, many more studies have been published at rapid pace, which makes the periodic review of prognostic knowledge essential.[2] This meta-analysis fills this gap by estimating the updated risk of transition to psychosis in individuals at CHR-P.

Key Points

Question What is the current likelihood of transitioning to psychosis in individuals at clinical high risk?

Findings In this meta-analysis of 130 longitudinal studies, including 9222 individuals at clinical high risk for psychosis from 74 cohorts, the risk of developing psychosis continued increasing after 2 years, cumulating to 25% at 3 years and reaching 35% at 10 years. Risk of transitioning to psychosis was higher in studies with a lower proportion of female individuals and a higher proportion of individuals presenting with brief limited intermittent psychotic symptoms.

Meaning This updated meta-analysis indicates that the probability of transitioning to psychosis in individuals at clinical high risk is substantial and continues increasing in the long term, suggesting that an extended duration of clinical monitoring and preventive care may be beneficial.

Methods

The study protocol was registered and made publicly available on the PROSPERO database (CRD42020168738) and followed the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) reporting guideline[15] (eTable 1 in the Supplement) and the Meta-analysis of Observational Studies in Epidemiology (MOOSE) reporting guideline[16] (eTable 2 in the Supplement).

Search Strategy and Selection Criteria

Three independent researchers (G.S.P., J.P., V.A.) performed a multistep literature search in PubMed and the Web of Science database (Clarivate Analytics), incorporating the Web of Science Core Collection, BIOSIS Citation Index, KCI-Korean Journal Database, MEDLINE, Russian Science Citation Index, and SciELO Citation Index, from inception until November 1, 2020 (eMethods 1 in the Supplement). We reviewed the references for previously published articles. We screened the titles and abstracts of the articles identified. We excluded those irrelevant and further assessed for inclusion the full texts of the remaining articles.

Studies were included if they (1) were original publications, (2) were conducted in individuals meeting CHR-P criteria (defined in eMethods 2 in the Supplement), (3) provided longitudinal data on transition to psychosis, and (4) were published in English. Studies were excluded if they were (1) reviews, clinical cases, conference proceedings, or protocols; (2) conducted in individuals not formally assessed with CHR-P instruments (eMethods 2 in the Supplement); or (3) cross-sectional. Randomized clinical trials were included using only the placebo or needs-based intervention arm. We contacted authors to gather missing data and to clarify overlaps. In case of overlapping samples reporting transition at the same follow-up time, we retained the largest and most recent sample. Agreement among the researchers was greater than 85% for study selection. We resolved disagreements through discussion and consensus with a senior academic (P.F-P.).

Measures and Data Extraction

At least 2 independent researchers (J.P., V.A., F.B., L.S., F.C., S.K., J.D.S.) extracted data, and a third independent researcher (G.S.P., A.C., J.V.S.) cross-checked the data. We extracted a predetermined set of variables (eMethods 3 in the Supplement) to describe the main characteristics of the studies or to conduct meta-regressions. The outcome variable we extracted was the raw number of individuals at CHR-P transitioning to psychosis at different follow-up times. We operationalized transition to psychosis as defined by each CHR-P instrument (eMethods 2 in the Supplement).

Quality Assessment

We evaluated the quality of the included cohorts using a modified version of the Newcastle-Ottawa Scale for cohort studies, which has been repeatedly used in the field.[6,17-19] We assessed the quality of randomized clinical trials with the Cochrane risk of bias tool[20] (eTable 3 and eMethods 4 in the Supplement).

Statistical Analysis

The primary effect size was the meta-analytical cumulative risk of transition to psychosis at 0.5, 1, 1.5, 2, 2.5, 3, 4, and more than 4 years' follow-up, estimated using the number of individuals at CHR-P transitioning to psychosis at each time point. To estimate the numbers of transitions, we used evidence from cohort studies using real-world electronic health records, indicating that transition risk is similar among individuals at CHR-P who were followed up with and those who were not followed up with (see Green et al[21] and eFigure 6 in the Supplement, published in Fusar-Poli et al[22]). This aligns with evidence showing that those who drop out share similar demographic characteristics,[23] similar functional impairments,[23] similar severity of attenuated psychotic

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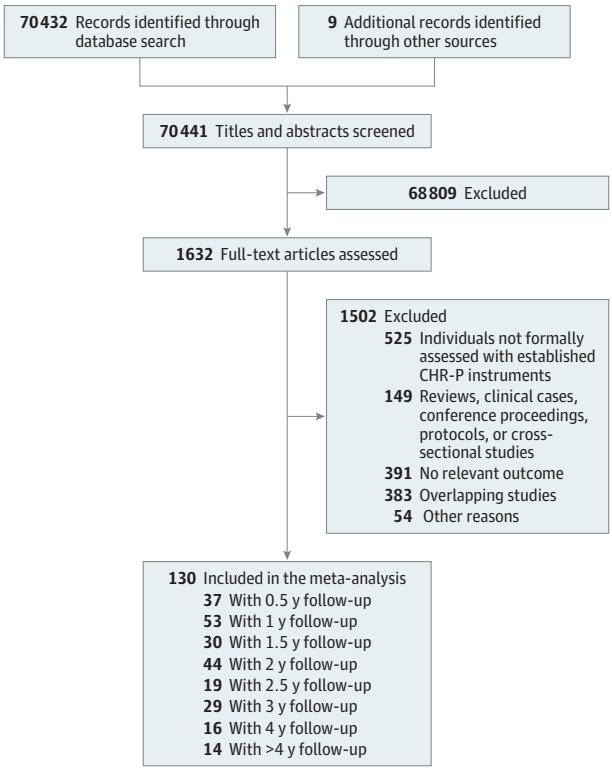
Figure 1. PRISMA Flowchart Outlining Study Selection Process



CHR-P indicates clinical high risk for psychosis.

symptoms[23] (or greater severity of disorganized symptoms,[24] which are associated with transition to psychosis[25]), and similar changes over time in the severity of symptoms.[23] We thus assumed an equal transition risk across the 2 groups and used the study-specific transition risk to compute the raw number of transitions among those who were not followed up with. Sensitivity analyses estimated the effect of such an assumption.

A second higher temporal granularity analysis was conducted within the subset of studies reporting a Kaplan-Meier survival curve; this analysis also accounted for dropouts. We digitally extracted the Kaplan-Meier curves from the articles included and estimated the time to transition (or end of follow-up) for each patient using a validated procedure (eMethods 5 in the Supplement).[26] Afterward, we used these individual data to compute the overall meta-analytical Kaplan-Meier failure curve (1 − survival). To further characterize the speed of transition to psychosis, a meta-analytic hazard rate curve of transition to psychosis was superimposed. Supplementary analyses (1) re-estimated the failure function starting 1, 2, 3, 4, and 5 years after the onset of the follow-up; (2) computed the frequency and percentage of transition over time; and (3) estimated the prediction intervals for the main analysis (eResults in the Supplement).

We performed multiple meta-regression analyses (ie, using 2 meta-regressor factors at the same time) of factors that were known to be associated with transition risk when at least 10 studies were available.[17,27] The fixed meta-regressor factor was follow-up time, which was combined with the following: publication year; study design; proportion of attenuated psychotic symptoms; brief intermittent psychotic symptoms (BIPS) or brief limited intermittent psychotic symptoms (BLIPS); genetic risk and deterioration syndrome; basic symptoms; mean age; proportion of female individuals; CHR-P assessment instrument; study quality; continent; duration of untreated attenuated psychotic symptoms; proportion of baseline comorbid mental disorders (according to any version of the *International Statistical Classification of Diseases and Related Health Problems* [*ICD*] or *DSM*); proportion of interventions at baseline; and follow-up. Meta-regression β coefficients were calculated to test how the outcome variable changed with a unit increase in the meta-regression factors.

Because the studies were expected to be heterogeneous, we used random-effects models. We assessed heterogeneity with the Q statistic and the I^2 index.[28] We tested for publication biases by conducting a Cox regression in which the dependent variable was the time to transition and the independent variable was the sample size. The Meta and Metaprop packages of Stata version 16 (StataCorp)[29] and the muhaz package for R version 1.2.6.3 (The R Foundation)[30] were used (eMethods 6 in the Supplement). All tests were 2-sided, and significance was set at $P < .05$.

Results

Sample Characteristics

The literature search yielded 70 441 citations, from which we assessed 1632 full-text articles. After excluding those not meeting the inclusion criteria, we included 130 studies from 74 cohorts in at least 1 of the follow-up meta-analyses (**Figure 1**). The overall database, considering all independent studies across all follow-up times, comprised 9222 individuals. The mean (SD) age was 20.3 (4.4) years, and 5100 individuals (55.3%) were male. Of 74 cohorts, most (36 [49%]) used the Structured Interview for Prodromal Syndromes (SIPS), followed by the Comprehensive Assessment of At-Risk Mental States (CAARMS; 29 [39%]), the Early Recognition Inventory based on the Interview for the Retrospective Assessment of the Onset of Schizophrenia (2 [3%]), the Positive and Negative Syndrome Scale (1 [1%]), and a combination of SIPS and CAARMS (1 [1%]) or SIPS and basic symptom-related instruments (5 [7%]). Most studies (37 [50%]) were carried out in Europe, followed by North America (19 [26%]), Asia (11 [15%]), Australia (4 [5%]), South America (1 [1%]), Africa (1 [1%]), and more than 1 continent (1 [1%]). The weighted mean (SD; range) duration of the follow-up was 37.4 (26.6; 3-192) months (eTable 4 in the Supplement).

Cumulative Risk of Transitioning to Psychosis in Individuals at CHR-P

The meta-analytical cumulative risk of transitioning to psychosis in individuals at CHR-P was 0.09 (95% CI, 0.07-0.10; k = 37; n = 6485) at 0.5 years, 0.15 (95% CI, 0.13-0.16; k = 53; n = 7907) at 1 year, 0.20 (95% CI, 0.17-0.22; k = 30; n = 5488) at 1.5 years, 0.19 (95% CI, 0.17-0.22; k = 44; n = 7351) at 2 years, 0.25 (95% CI, 0.21-0.29; k = 19; n = 3114) at 2.5 years, 0.25 (95% CI, 0.22-

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0.29; k = 29; n = 4029) at 3 years, 0.27 (95% CI, 0.23-0.30; k = 16; n = 2926) at 4 years, and 0.28 (95% CI, 0.20-0.37; k = 14; n = 2301) at more than 4 years (**Figure 2**; **Table 1**). The sensitivity analyses are reported in eTable 5 and eFigure 1 in the Supplement. Prediction intervals and assessment of publication bias are reported in the eResults in the Supplement.

Meta-analytical Kaplan-Meier Failure and Hazard Rate of Transitioning to Psychosis in Individuals at CHR-P

A subset of studies (k = 25) plotted Kaplan-Meier estimates of transition risks. The meta-analytical failure function is summarized in **Figure 3**. The cumulative estimates of transition (**Table 2**) were 0.08 (95% CI, 0.08-0.09; n = 4860) at 0.5 years, 0.14 (95% CI, 0.13-0.15; n = 3408) at 1 year, 0.17 (95% CI, 0.16-0.19; n = 2892) at 1.5 years, 0.20 (95% CI, 0.19-0.21; n = 2357) at 2 years, 0.25 (95% CI, 0.23-0.26; n = 1444) at 2.5 years, 0.27 (95% CI, 0.25-0.28; n = 1029) at 3 years, 0.28 (95% CI, 0.26-0.29; n = 808) at 3.5 years, and 0.29 (95% CI, 0.27-0.30; n = 737) at 4 years. However, the transition risk increased to 0.35 (95% CI, 0.32-0.38; n = 114) at 10 years (Table 2). The speed of transition to psychosis declined from 0.14 (95% CI, 0.13-0.15) at 0.5 years to 0.08 (95% CI, 0.07-0.09) at 2 years, to 0.05 (95% CI, 0.04-0.06) at 3 years, and to 0.02 (95% CI, 0.02-0.04) at 4 years when it plateaued (Figure 3; eTable 6 in the Supplement). The survival curves re-estimated at 1 to 5 years from the start of follow-up are appended in eTables 7 to 11 in the Supplement. The frequency and percentage of transitions over time are appended in eFigure 2 in the Supplement.

Heterogeneity and Quality Assessment

The mean (SD; range) Newcastle-Ottawa Scale score was 4.5 (0.9; 3-7) (eTable 4 in the Supplement). A high risk of bias was found in 50% of randomized clinical trials. Heterogeneity across the studies was statistically significant (Q range, 79.12 to 304.36; I^2 range, 77.91% to 95.73%; $P < .001$) in the main meta-analysis.

Meta-regressions

Multiple meta-regression analyses were controlled for follow-up time, which was associated with transition to psychosis (β = 0.002; 95% CI, 0.001-0.003; z score = 3.93; $P < .001$). Risk of transitioning to psychosis was higher in studies with a lower proportion of female individuals (β = −0.02; 95% CI, −0.04 to −0.01; z score = −2.83; $P = .005$). Having a higher proportion of BLIPS/BIPS was associated with higher transition risk (β = 0.02; 95% CI, 0.01-0.03; z score = 2.71; $P = .007$) (eTable 12 in the Supplement). There were no significant associations between any other evaluated meta-regressors and transition to psychosis (eTable 12 in the Supplement).

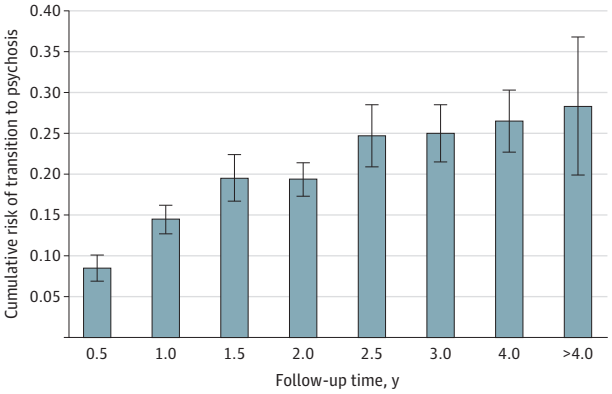
Discussion

This updated meta-analysis showed that about one-fourth of individuals at CHR-P developed psychosis within 2 to 3 years but that their transition risk continued to increase in the long term. To our best knowledge, with 130 studies and up to 9222 individuals at CHR-P, this is the largest meta-analysis addressing the likelihood of transitioning to psychosis. The current database is globally representative, including individuals at CHR-P from Europe (50%), North America (26%), Asia (15%), Australia (5%), South America (1%), and Africa (1%). This distribution matches the global clinical CHR-P research conducted over the past 2 decades, including more than 20 000 young people (Europe, 51%; North America, 17%; East Asia, 17%; Australia, 6%; South America, 6%; and Africa, 2%[31]). These findings are particularly relevant in the context of the substantial sampling biases that affect the CHR-P paradigm,[32,33] which lead to substantial risk enrichment during the recruitment of young individuals undergoing CHR-P assessment.[22,34] The reported transition risks are transitional in nature, cutting across adult and children or adolescent populations. While CHR-P criteria are typically validated for those aged 12 to 35 years,[10] the current mean age was around 20 years, indicating that the CHR-P paradigm is more geared toward adolescents and young adults. We also confirmed that most individuals at CHR-P were male (55.3%), in line with the epidemiological distribution of psychosis risk across sexes.[3,5,18]

From a statistical perspective, the sample size included in this meta-analysis is comparable with that of other meta-analyses of clinical high-risk syndromes in somatic medicine (44 studies in mild cognitive impairment to dementia[35] and 103 studies in intermediate hyperglycemia to diabetes[36]). This translates into high statistical power and improved precision, with meta-analytical confidence intervals (Table 1) of about 10% (at 1 and 10 years), similar to those observed in meta-analyses of intermediate hyperglycemia to diabetes (eg, confidence intervals of about 62% at 1 year to about 41% at 10 years[36]).

The first main finding of this study is to substantiate a consistent increase in likelihood of developing psychosis across CHR-P samples, despite their clinical heterogeneity.[37-39] Individuals at CHR-P have a substantially higher risk of developing psychosis (0.15 at 1 year) than the general population (global incidence, 26.6 per 100 000 person-years; 95% CI, 22.0-31.7[40,41]). This finding tempers recent controversies in this field,[42] advocating for indicated preventive efforts in this

Figure 2. Meta-analytic Cumulative Risk of Transition to Psychosis in Individuals at Clinical High Risk for Psychosis



Error bars indicate 95% CIs.

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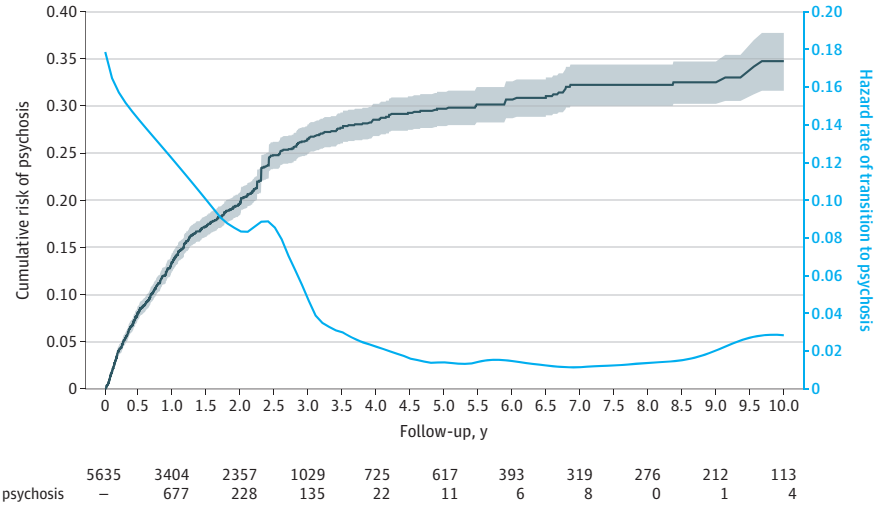
Table 1. Meta-analytic Cumulative Risk of Transition to Psychosis in Individuals at Clinical High Risk for Psychosis

Follow-up, y	Studies, No.	No. of transitions to psychosis	Sample size	Meta-analytical cumulative risk of transition to psychosis (95% CI)	Q	*df*	I^2
0.5	37	595	6485	0.09 (0.07-0.10)	174.65	36	79.39
1.0	53	1236	7907	0.15 (0.13-0.16)	235.36	52	77.91
1.5	30	1168	5488	0.20 (0.17-0.22)	195.34	29	85.15
2.0	44	1497	7351	0.19 (0.17-0.22)	197.56	43	78.23
2.5	19	791	3114	0.25 (0.21-0.29)	107.81	18	83.30
3.0	29	1080	4029	0.25 (0.22-0.29)	183.24	28	84.72
4.0	16	903	2926	0.26 (0.23-0.30)	79.12	15	81.04
>4.0	14	755	2301	0.28 (0.20-0.37)	304.36	13	95.73

Figure 3. Meta-analytical Kaplan-Meier Failure Function (1 − Survival) and Hazard Rate of Transition to Psychosis in Individuals at Clinical High Risk for Psychosis



No. at risk	5635	3404	2357	1029	725	617	393	319	276	212	113
No. transitions to psychosis	–	677	228	135	22	11	6	8	0	1	4

The shaded area indicates 95% CIs.

vulnerable population.[1] Recent studies have confirmed that transition to psychosis is a clinically meaningful end point for individuals at CHR-P, which translates into real-world morbidity.[12,43]

This study also specifically quantifies the level of transition risk, which increased from 0.09 at 0.5 years to 0.25 at 3 years to 0.27 at 4 years. Importantly, our main analyses, assuming equal transition risk across those who dropped out and those who did not (0.27 at 4 years), were comparable with Kaplan-Meier–based meta-analytical estimates accounting for dropouts (0.29 at 4 years). These findings support electronic health records as useful monitoring tools in individuals at CHR-P[44,45] and the use of more assertive follow-up with those not engaging in future clinical research studies.[21] Overall, the levels of risk enrichment, and conversely of false-positives, are comparable with the likelihood of developing diabetes from impaired glucose tolerance (0.13 at 1 year, 0.16 at 2 years, and 0.22 at 4 years), with regression to normoglycemia (ie, false-positive results) observed in up to 80% of patients.[36] Furthermore, most of those individuals at CHR-P not transitioning to psychosis did not recover but continued displaying persistent symptoms, comorbidities, and high clinical needs.[46-51]

These updated meta-analytical findings may guide prognostication efforts in clinics for individuals at CHR-P. Some individualized risk calculators have been developed and validated,[52-58] but, to our knowledge, none have yet been implemented in clinical routine.[59] For example, our risk estimates can be shared during the nuanced prognostic communication with young people at CHR-P and their caregivers (if requested), balancing their uncertainty and the ethical imperatives of nonmaleficence vs beneficence and autonomy.[60] From a research perspective, our updated estimates can inform power calculations[61] for future international consortia exploring prognostic models or novel interventions.

The second main finding of the current study is that the risk of transition to psychosis continued increasing up to 0.35 at 10 years. To our best knowledge, this is the first meta-analysis to have explored the long-term course of psychosis risk in individuals at CHR-P and the first to have estimated the long-term speed of transition to psychosis. The rate of transition to psychosis was highest in the first months after the ini-

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Table 2. Meta-analytical Estimates of the Kaplan-Meier Failure Function (1 − Survival) of Transition to Psychosis in Individuals at Clinical High Risk for Psychosis

Follow-up time, y	No. at risk	Cumulative No. of transitions to psychosis	Meta-analytical cumulative risk of transition to psychosis (95% CI)
0.5	4860	451	0.08 (0.08-0.09)
1.0	3408	677	0.14 (0.13-0.15)
1.5	2892	819	0.17 (0.16-0.19)
2.0	2357	905	0.20 (0.19-0.21)
2.5	1444	1013	0.25 (0.23-0.26)
3.0	1029	1040	0.27 (0.25-0.28)
3.5	808	1053	0.28 (0.26-0.29)
4.0	737	1062	0.29 (0.27-0.30)
4.5	662	1069	0.29 (0.27-0.31)
5.0	628	1073	0.30 (0.28-0.32)
5.5	420	1076	0.30 (0.28-0.32)
6.0	397	1079	0.31 (0.29-0.34)
6.5	373	1081	0.31 (0.33-0.29)
7.0	323	1087	0.31 (0.30-0.34)
7.5	323	1087	0.31 (0.30-0.34)
8.0	323	1087	0.31 (0.30-0.34)
8.5	250	1088	0.32 (0.30-0.35)
9.0	250	1088	0.32 (0.30-0.35)
9.5	132	1089	0.32 (0.30-0.35)
10.0	114	1092	0.35 (0.32-0.38)

tial CHR-P assessment,[62] with about 80% of transitions occurring within 2 years (eFigure 2 in the Supplement). Preventive care should be promptly implemented by CHR-P services and the use of waiting lists or watchful waiting strategies discouraged. After these initial months, the rate of transition to psychosis declined, but it did not plateau until about 4 years (with 190 transitions occurring after 2 years; eFigure 2 in the Supplement). This finding contrasts with the mean duration of care offered by most CHR-P services worldwide, which is generally capped at 2 years or less,[10] irrespective of clinical need (only 15% of individuals who were at CHR-P at baseline eventually remit[46]) or subjective preference (that is, individuals at CHR-P are discharged at 2 years even if they would prefer to remain in CHR-P care). Notably, Figure 3 depicts a slight increase in the rate of transition just past the 2-year discharge point, suggesting a rebound increase of transition risk postdischarge. The probability of individuals at CHR-P developing other severe real-world outcomes similarly doubles from the short term to the long term: risk of informal hospital admission increases from 0.12 at 2 years to 0.32 at 10 years and risk of compulsory admission increases from 0.08 at 2 years to 0.25 at 10 years.[44] A similar increase in psychotropic treatments in the long term is observed.[44] The accumulated long-term clinical burden leads to a concerning 9-year mortality ratio (standardized for age and sex) of 3.9.[44] Overall, our meta-analysis suggests that the duration of care currently offered by CHR-P services may be too short to capture these long-term outcomes and could, therefore, be extended to a minimum of 4 years, at least for individuals experiencing persistent disability at 2-year follow-up and according to their preference. Future research should also develop and validate stratification models to identify those individuals more likely to benefit from a short vs long duration of care in CHR-P services.

The third main finding of the current study is to have helped deconstruct heterogeneity in transition risk by comprehensively testing numerous meta-regressor factors. Transition risks were higher in studies with a lower proportion of female individuals and in those with a higher proportion of individuals presenting with BLIPS/BIPS, in line with recent large-scale studies.[63] Umbrella reviews have consolidated male sex as an established prognostic factor for psychosis (incidence rate ratio for male individuals vs female individuals, 1.34[5]). The presence of short-lived psychotic episodes has consistently been associated with a very high risk of transition (0.51; 95% CI, 0.41-0.61 at 36 months or more[64]), in particular if recurrent (hazard ratio compared with single episodes, 3.98[25]) or associated with seriously disorganizing or dangerous features (0.90 at 5 years; 95% CI, 0.64-0.99[25]). Individuals with BLIPS/BIPS had a peculiar clinical presentation (35% with abrupt onset, 32% with acute stress, 45% with lifetime trauma, and 20% with illicit substance use) and clinical outcomes beyond psychosis[65] that were often severe (34% were admitted to the hospital and 16% were compulsorily admitted to the hospital at 4 years) and unmet by current interventions (only a minority received the appropriate dose of cognitive behavioral therapy[66]). Age did not moderate transition risk, in line with recent meta-analyses showing commensurable risks in adults as well as children and adolescents at CHR-P.[19] The type of CHR-P instrument or their combination with basic symptom-based instruments and the presence of baseline nonpsychotic comorbid disorders did not affect transition risks either, in line with previous meta-analyses[11,67,68] and original

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studies.[69] There were no differences across continents, supporting the global scalability of the CHR-P paradigm.[31] We did not confirm the declining transition risk over the most recent years, which we first observed meta-analytically.[14] After an early drop, transition risk may have remained relatively stable more recently. This phenomenon has been interpreted via a dilution effect,[70] mediated by recruitment procedures,[32] or by a treatment effect, mediated by preventive interventions.[71] However, there is no convincing evidence for specific preventive interventions reducing the transition risk[72,73]; our meta-regression analyses confirmed that exposure to antipsychotics, antidepressants, or psychotherapy did not moderate transition risk.

Limitations

This study has some limitations. First, there were too few primary studies focusing on basic symptoms. Second, we did not explore outcomes other than psychosis,[74,75] because this would have required a different meta-analytic approach. Third, there was high heterogeneity; we accounted for it in meta-regression analyses but other significant sources of heterogeneity may exist. Fourth, attrition was variable across cohorts (eDiscussion in the Supplement); this issue was dealt with through different types of analyses. Fifth, the examined independent variables were group means, and their analyses might have been subject to ecological bias. Sixth, most studies referred to transition or conversion to psychosis only and the specific *ICD* or *DSM* psychotic disorder diagnoses met by individuals who transitioned were infrequently reported by the original studies. Seventh, there were not enough primary studies reporting outcomes at 3.5 years. Eighth, we could not conduct meta-regressions for the duration of untreated CHR-P[76] or baseline severity of attenuated psychotic symptoms, as limited studies provided these data with heterogeneous instruments. We could not conduct meta-regressions for some of the baseline comorbid mental disorders and baseline and follow-up interventions for the same reason (eTable 12 in the Supplement). Ninth, although we suggested extending duration of care in CHR-P services until 4 years, the effectiveness of this approach should be confirmed by future randomized studies comparing long vs short duration of care provided by CHR-P services. Similar studies, although logistically and empirically challenging, have been conducted in patients with a first episode of psychosis[77] and may become a mainstream field of research in the near future.

Conclusions

In this meta-analysis, 25% of individuals at CHR-P developed psychosis at 3 years. Their transition risk to psychosis continued increasing in the long term. Extended clinical monitoring and preventive care may be beneficial in this patient group.

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